

# LAVYA MIDHA

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## EDUCATION

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### New York University

Master of Science in Data Science (Industry Concentration)

Cumulative GPA: 3.815/4.0

Relevant Coursework: Probability, Statistics, Text as Data, Big Data, Machine Learning, Practical Training for DS, ML in Finance.

New York, NY

Expected May 2026

### Boston University

Bachelor of Science in Data Science, Minor: Statistics

Cumulative GPA: 3.75/4.0

Relevant Coursework: Data Visualization, Machine Learning and AI, Algorithms, Statistical Modeling, Data Mechanics, Optimization, NLP, Regression Analysis, Inference, Analysis of Variance.

Boston, MA

Sept 2021 - May 2024

## WORK EXPERIENCE

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### ReferU.AI

Machine Learning Engineer Intern

Lewes, Delaware

May 2025 – Present

- **End-to-end GenAI Legal Assistant:** Designed and deployed a production-grade, multi-agent RAG system (LangChain + OpenAI + GCP) for citation-aware legal retrieval, integrating retrieval → reranking → guardrail pipelines with Dockerized FastAPI microservices and CI/CD automation.
- **Retrieval Scoring Framework:** Built a scalable benchmarking and evaluation pipeline for court opinion evaluation (GCP + CourtListener API) to measure precision/recall, latency, and coverage; driving data-driven optimization across 10K+ legal citations.
- **Adversarial Validation Layer:** Engineered a parallel multi-agent QA framework (Validator, Auditor, Bias Hunter, Edge-Case Challenger) to reduce hallucinations and bias, boosting factual accuracy +22% and cutting manual review effort 35%

### BU Spark!

Project Manager Intern

Boston, MA

Jan 2024 – May 2024

- Directed 3 applied ML teams delivering forecasting and optimization tools for civic-tech clients; acted as the primary client liaison, translating non-technical policy goals into ML problem statements and deployment plans.
- Collaborated with teams to build Tableau dashboards that converted complex ML outputs into intuitive, non-technical insights for city officials and executives, enabling data-driven decision-making.
- Improved iteration speed by 14% by introducing agile pipelines and structured stakeholder review loops, ensuring deployed models were interpretable, accurate, and aligned with end-user needs.

### Boston University School of Public Health

Data Science Research Intern

Boston, MA

Feb 2023 - Dec 2023

- Informed \$MM+ infrastructure investments by building time-series models on air quality data from 125+ schools, prioritizing equity-first capital allocation.
- Surfaced systemic disparities in classroom environments by designing an OpenCV pipeline to extract spatial features (window access, density) from floor plans.
- Cut anomaly response times by 35% by deploying a real-time dashboard flagging temperature spikes for administrators and facilities teams.

## SELECTED PROJECTS

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- **Probability of Default Prediction System:** Created an end-to-end PD modeling system for SME credit risk, integrating extensive ratio engineering, XGBoost modeling, and advanced calibration (Laplace, bucket mapping) to produce accurate, well-behaved PDs validated via OOT splits and AUC/KS/Brier diagnostics.
- **Voice PrepAI:** Developed an LLM-driven interview preparation agent with real-time audio streaming and multi-agent evaluation that provides personalized, resume-, role-, and company-aware feedback to improve answer quality and structure.
- **Bitcoin Accumulation (for Trilemma Foundation):** Built a multi-agent system for bitcoin accumulation in which learning-based agents detect market regimes, forecast short-term risk, and adjust DCA schedules in real time to maximize accumulated BTC across varying market conditions.

## SKILLS

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**Programming & Data:** Python (Pandas, NumPy, scikit-learn, matplotlib) , SQL, C++, Java, R, TypeScript, Node.js.

**ML/ AI:** PyTorch, TensorFlow, LightGBM, XGBoost, Transformers, Recommender Systems, Time Series, NLP, Computer Vision, GenAI & Multi-Agent Systems, LangChain.

**Data Infrastructure:** Spark, Hadoop ,Dataproc, Docker ,REST APIs, ETL, HPC, JIRA.

**Cloud & Deployment:** GCP, AWS, Azure, Google Ads, Google Analytics ; CI/CD, Monitoring.

**Visualization & Communication:** Tableau, PowerBI, Looker Studio Python Panel, Rshiny Stakeholder-Facing, Agile.